IQBAL GROUP OF BS ZOOLOGY
03460004097

ZOO202 PAST PAPERS OF 2022

اپنا لکھا مجھے معیوب لگے
وقت لکھے گا تعارف میرا

بسم اللہ الرحمٰن

الرحیم

یااللہ یارسول ﷺ مدد

کریم مدد

لبیک یا رسول اللہ

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Tlp zindabad Pakistan zindabad Islam zindabad Pak army zindabad zindabad

وہ ﷺ اتنا دیتے ہیں کہ جھولیاں بھر دیتے ہیں

میرے آقا کے جودو سخا پہ لاکھوں درودوسلام

یہ کہتی ھے بلبل باغ جِناں نہیں کوئی رضا کی طرح سہربیاں

نہیں ہند میں واصف شاہ ہدیٰ مجھے شوخی طبع رضا کی قسم

(اعلٰحضرت امام احمد رضا خان بریلوی)

اقبال گروپ آف بی ایس زوالوجی

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OBJECTIVE

Our mission and moto is help to others and develop the Muslims ,our nation and our country. May Allah pak grant us the power to complete our mission.

Regards : Iqbal group of bs zoology

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EDUCATION

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وہ اتنا دیتے ہیں کہ جھولیاں بھر دیتے ہیں
میرے آقاﷺ کے جودو سخا پہ لاکھوں کروڑوں درودوسلام

Qno1 Define cochlea?

- The cochlea is a hollow, spiral-shaped bone found in the inner ear that plays a key role in the sense of hearing and participates in the process of auditory transduction. Sound waves are transduced into electrical impulses that the brain can interpret as individual frequencies of sound.

Qno2 write three names of pelvic girdle of amphibians?

- The pelvic girdle, also known as the hip bone, is composed of three fused bones: the ilium, ischium and the pubic bone. The pelvic spine is the posterior portion of the pelvis below the lumbar spine, composed of the sacrum and coccyx.

- **Qno3 write the name of ancient bird 🕊️ found in china?**

Gansus yumenensis is the first Mesozoic bird found in China. It was identified following the discovery of a fossil near Changma, Gansu Province, northwestern China in 1984.

Qno4 Define non-shevering thermogenesis?

- Nonshivering thermogenesis is defined as an increase in metabolic heat production (above the basal metabolism) that is not associated with muscle activity. It refers mainly to the increased metabolism of brown fat, but to a lesser degree can also be detected in skeletal muscles, liver, brain, and white fat.

Qno5 Write difference between crocodile 🐊and alligator?

- **Alligator and crocodile**

  The main thing to remember is that alligators and crocodiles are different species. They can be identified by their jaw shape, size and color. Alligators are dark green or black, have u-shaped snouts, tiny black spots and are smaller than crocs. Crocodiles are bigger, have v-shaped snouts and are more aggressive.

Qno6 Difference between proventriculus and ventriculus?

- As nouns the difference between proventriculus and ventriculus is that proventriculus is the part of the avian stomach, between the crop and the gizzard, that secretes digestive enzymes while ventriculus is a digestive cavity or stomach.

Qno7 Write the characteristics of mammals?

- Characteristics of Mammals
- The presence of hair or fur.
- Sweat glands.
- Glands specialized to produce milk, known as mammary glands.

- Three middle ear bones.
- A neocortex region in the brain, which specializes in seeing and hearing.
- Specialized teeth.

A four-chambered heart.

Qno8 Write a note on negative pressure mechanism in reptiles?

- Negative pressure mechanism

Most reptiles breathe by changing the volume of the body cavity. By contractions of the muscles moving the ribs, the volume of the body cavity is increased, creating a negative pressure, which is restored to atmospheric level by air rushing into the lungs.

Qno9 Write the nervous system of enteropneustsis?

- Nervous system
- The nervous system of enteropneustsis ectodermal in origin
- Lies at the base of the ciliated epidermis.
- Consists of dorsal and ventral nerve tracts
- and a network of epidermal nerve cells, called a nerve plexus.
- There are no major ganglia.

Sensory receptors are unspecialized and widely distributed over the body.

Qno10 Define clutch?

The group of eggs laid and chicks produced by a female is called a clutch.

Qno11 Explain pheromones?

- Pheromones are used to recognize members of
- the same species, members of the opposite sex,
- and the reproductive state of a member of the

- opposite sex
- • Pheromones may also induce sexual behavior,
- help establish and recognize territories, and ward

  off predators.

Qno12 What is Hibernacula?

- Individuals that are usually solitary may migrate to a common site to spend the

  winter. Heat loss from these groups, called hibernacula.

Qno13 What is pygostyle?

- The posterior caudal vertebrae are fused into a pygostyle, which helps

  support the tail feathers that are important in steering.

Qno14 Ram ventilation?

Ram ventilation constitutes swimming in relatively straight lines at a high rate of speed with the mouth open. This forces water through the mouth and over the gills, meaning the fish can breathe passively while swimming. Some classic examples are tuna, swordfish and shark.

Qno15 Differentiate between Homodont and hetrodont?

Homodont - Teeth are all about the same shape (most vertebrates, few mammals). Heterodont - Teeth have different form and functions in different parts of the tooth row (mammals, a few fish).

Qno16 What is siphons?

- The adult tunicate
- body is shaped like a small barrel with two

  large openings called siphons.

Qno17 What is secondary palate?

- Secondary palate
- A plate of bone, called the secondary palate, evolved in the archosaurs and separates

  the nasal and mouth passage ways.

Qno18 Define lateral line system?

- The lateral-line system consists of sensory pits in the epidermis In these
- pits are receptors that are stimulated by water

  • Lateral lines are used to detect either water currents, or apredator.

Qno19 explain countercurrent mechanism?

- Countercurrent heat-exchange systems
- may help
- regulate heat loss from exposed areas (figure 22.14)
- • Arteries passing peripherally through the core of an
- appendage are surrounded by veins that carry blood
- back toward the body
- • When blood returns to the body through these veins,
- heat transfers from arterial blood to venous blood and
- returns to the body rather than being lost to the

  environment.

Qno20 Explain types of flight?

- different kinds of flight
-

- 1• During gliding flight, the wing is stationary, and a bird loses altitude. Waterfowl coming in for a
- landing use gliding flight
- 2• Flapping flight generates the power for flight and is the most common type of flying. Many
- variations in wing shape and flapping patterns result in species-specific speed and
- maneuverability
- 3• Soaring flight allows some birds to remain airborne with little energy expenditure. During
- soaring, wings are essentially stationary, and the bird utilizes updrafts and air currents to gain
- altitude.
- 4 . hovering flight.
- They hover in still air by fanning their wings back and forth.
-

Qno21 Describe Types of layers on the reptilian egg?

- Amniotic egg consists of three layers, The amnion encloses the embryo in a fluid-filled
- sac and protects against shock and desiccation. The chorion is nearer the shell and
- becomes highly vascular and aids in gas exchange. The allantois is a ventral outgrowth

of the gut and stores nitrogenous wastes (e.g., uric acid).

Qno22 What is Embryonic Diapause?

- Development is arrested for the three months during which the recently born young do

  most of their nursing. This mechanism is called embryonic diapause.

Qno23 What is hibernation?

- Hibernation is a period of winter inactivity in which the hypothalamus of the brain slows
- the metabolic, heart, and respiratory rates
- • True hibernators include the monotremes (echidna and duck-billed platypus) and many
- members of the Insectivora (e.g., moles and shrews), Rodentia (e.g., chipmunks and
- woodchucks), and Chiroptera (bats)

  • In preparation for hibernation, mammals usually accumulate large quantities of body fat.

- **Qno24 Differentiate between Shivering and Non-shivering thermo genesis?**
- Shivering thermogenesis is muscular activity that generates large amounts of heat but
- little movement.
- Nonshivering thermogenesis involves heat production by general cellular metabolism
- and the metabolism of special fat deposits called brown fat
- Heat production is effective in thermoregulation because mammals are insulated by their
- pelage and/or fat deposits

Fat deposits are also sources of energy to sustain high metabolic rates.

- **Qno25 What is Convergent evolution?**

- The evolution of similar structures in different lineages is called convergent evolution
- The mammalian heart evolved in the synapsid reptilian lineage, whereas the avian heart evolved

in the archosaur lineage.

- **Qno26 Differentiate between Bronchi and parabronchi?**

- Bone and cartilage support the trachea
- A special voice box, called the syrinx, is located where the trachea divides into bronchi.
- The muscles of the syrinx and bronchi, as well as characteristics of the trachea, produce
- bird vocalizations
- The bronchi lead to a complex system of air sacs that occupy much of the body and
- extend into some of the bones of the skeletal system (figure 21.10a)
- The air sacs and bronchi connect to the lungs. The lungs of birds are made of small air

tubes called parabronchi**.**

- **Qno27 What is preening?**

- Birds maintain a clean plumage to rid the feathers and skin of parasites. Preening, which
- is done by rubbing the bill over the feathers, keeps the feathers smooth, clean, and in

- place. Hamuli that become dislodged can be rehooked by running a feather through the

  bill.

Long question

- **Qno28 Describe the Types of temporal region opening in reptiles?**
- • There are three classes of reptiles according to opening in temporal region
- • Reptiles in the subclass Anapsida lack openings or fenestrae region of the skull. The
- turtles represent this lineage today.
- • A second group of reptiles are diapsid. They have upper and lower openings in the
- temporal region of the skull.
- • Third group of reptiles are synapsid. They possess a single dorsal opening in the temporal
- region of the skull.
- Cladistic analysis shows that the birds (traditionally the class Aves) and the mammals

  (traditionally the class Mammalia) share a common ancestor with the reptiles.

- **Qno29 Explain Order of reptiles?**
- ● They have dry skin with epidermal scales
- • Skull with one point of articulation with the vertebral column (occipital condoyle);
- respiration via

- • Lungs are present in reptiles
- • Osmoregulation occurs through metanephric kidneys
- • internal fertilization
- • Amniotic eggs
- • They have four orders named as; 1. **Testudines**, 2. **Rhynchocephalia** 3. **Squamata**, 4.
- **Crocodilia**
- Order testudines (Chelonia)
- · Members of the order Testudines or Chelonia are the turtles
- · The approximately 225 species of turtles are characterized by a bony shell, limbs
- articulating internally to the ribs, and a keratinized beak rather than teeth.
- · The dorsal portion of the shell is the carapace, which forms from a fusion of
- vertebrae, expanded ribs, and bones in the dermis of the skin. Keratin covers the
- bone of the carapace.
- · The ventral portion of the shell is the plastron. It forms from bones of the
- pectoral girdle and dermal bone, and keratin also covers it
- Order rhynchocephalia
- • The one surviving species of the order is the tuatara (Sphenodon punctatus)
- • Tooth attachment and structure distinguish the tuatara from other reptiles.

- • It is oviparous and shares under- ground burrows with ground-nesting seabirds
- • Tuataras venture out of their burrows at dusk and dawn to feed on insects or,
- occasionally, small vertebrates.
- Order Squamata
- Order Squamata divided into three suborders.
- **1. Suborder Sauria—The Lizards**
- About 3,300 species of lizards are in the suborder Sauria
- Lizards usually have two pairs of legs, and their upper and lower jaws unite anteriorly
- Lizards vary in length from only a few centimeters to as large as 3 m. Many lizards live
- on surface substrates and retreat under rocks or logs when necessary
- Others are burrowers or tree dwellers.
- **2. Suborder Serpents—The Snakes**
- About 2,300 species are in the suborder Serpents
- Three hundred species are venomous
- Snakes are elongate and lack limbs
- The skeleton may contain more than two hundred vertebrae and pairs of ribs
- Joints between vertebrae make the body very flexible
- Snakes possess skull adaptations that facilitate swallowing large prey

- Other differences between lizards and snakes include the mechanism for focusing the
- eyes and the morphology of the retina.
- Most snakes are oviparous, although a few, such as the New World boas and garter
- snakes, give birth to live young.
- **3. Suborder Amphisbaenia—Worm-Lizards**
- About 135 species are in the suborder Amphisbaenia. They are specialized burrowers that live in soils of Africa, South America, the
- Caribbean, and the Mideast
- Most are legless, and their skulls are wedge or shovel shaped. A single median tooth in
- the upper jaw distinguishes amphisbaenians from all other vertebrates.
- The skin of amphisbaenians has ring like folds called annuli.
- Muscles of the skin cause it to telescope and bulge outward, forming an anchor against a
- burrow wall
- Amphisbaenians move easily forward or backward—thus, the suborder name. They feed
- on worms and small insects and are oviparous.
- **4. Order Crocodilia**
- The order Crocodilia has 21 species
- Along with dinosaurs, crocodilians are derived from the archosaurs and distinguished

- from other reptiles by certain skull characteristics: openings in the skull in front of the
- eye, triangular rather than circular eye orbits, and laterally compressed teeth
- Living crocodilians include the alligators, crocodiles, gavials, and caimans.
- Crocodilians have not changed much over their 170 million-year history
- The snout is elongate and often used to capture food
- The nostrils are at the tips of the snout, so the animal can breathe while mostly
- submerged
- Air passageways of the head lead to the rear of the mouth and throat, and a flap of tissue
- near the back of the tongue forms a watertight seal that allows breathing without inhaling

  water in the mouth.

- **Qno30 Counter current heat exchange mechanism?**
- ● Countercurrent heat-exchange systems may help regulate heat loss from exposed areas
- • Arteries passing peripherally through the core of an appendage are surrounded by veins
- that carry blood back toward the body
- • When blood returns to the body through these veins, heat transfers from arterial blood to
- venous blood and returns to the body rather than being lost to the environment

- • When excess heat is produced, blood is shunted away from the countercurrent veins
- toward peripheral vessels, and excess heat is radiated to the environment
- • Mammals have few problems getting rid of excess heat in cool, moist environments
- • Heat can be radiated into the air from vessels near the surface of the skin or lost by
- evaporative cooling from either sweat glands or respiratory surfaces during panting
- • Hot, dry environments present far greater problems, because evaporative cooling may
- upset water balances.
- • Jackrabbits and elephants use their long ears to radiate heat. Small mammals often avoid
- the heat by remaining in burrows during the day and foraging for food at night

• Other mammals seek shade or watering holes for cooling.

- **Qno31 Explain Inspiration and expiration in mammals?**
- • Mammalian lungs, like those of reptiles, inflate using a negative-pressure mechanism
- • Unlike reptiles and birds, however, mammals possess a muscular diaphragm that eparates the thoracic and abdominal cavities
- • Inspiration results from contraction of the diaphragm and expansion of the rib cage, both

- of which decrease the intrathoracic pressure and allow air to enter the lungs
- • Expiration is normally by elastic recoil of the lungs and relaxation of inspirator muscles,
- which decreases the volume of the thoracic cavity
- • The contraction of other thoracic and abdominal muscles can produce forceful exhalation.
- **Qno32 Explain External structure of reptiles?**
- ● Reptilian skin is thick, dry, and keratinized
- • Scales may be modified for various functions. For example, the large belly scales of
- snakes provide contact with the substrate during locomotion
- • Skin secretions include pheromones that function in sex recognition and defense
- • All reptiles periodically shed the outer, epidermal layers of the skin in a process called
- ecdysis
- • The frequency of ecdysis varies from one species to another, and it is greater in juveniles
- than adults
- • Cryptic coloration, mimicry, and aposematic coloration occur in reptiles. Color and color
- change also function in sex recognition and thermoregulation
- • Primitive reptiles move by crawling on earth surface

- • Many prehistoric reptiles were bipedal, meaning that they walked on hind limbs
- • They had a narrow pelvis and a heavy, outstretched tail for balance
- • Bipedal locomotion freed the front appendages, which became adapted for prey capture

  or flight in some animals.

- **Qno33 Explain Types of Flights in birds?**
- Different birds, or the same bird at different times, use different kinds of flight
- During gliding flight, the wing is stationary, and a bird loses altitude. Waterfowl coming
- in for a landing use gliding flight
- Flapping flight generates the power for flight and is the most common type of flying.
- Many variations in wing shape and flapping patterns result in species-specific speed and
- maneuverability
- Soaring flight allows some birds to remain airborne with little energy expenditure.
- During soaring, wings are essentially stationary, and the bird utilizes updrafts and air
- currents to gain altitude. Hawks, vultures, and other soaring birds are frequently observed
- circling along mountain valleys, soaring down-wind to pick up speed and then turning

- upwind to gain altitude.
- Hummingbirds perform hovering flight. They hover in still air by fanning their wings
- back and forth (50 to 80 beats per second) to remain suspended in front of a flower or

  feeding station.
- **Qno34 Explain Teeth Types in mammals?**
-
- • The first teeth emerge before or shortly after birth and are called deciduous or milk
- teeth
- • These teeth are lost, and permanent teeth replace them
- • Adult mammals have up to four kinds of teeth
- • 1. Incisors are the most anterior teeth in the jaw. They are usually chisel-like and used
- for gnawing or nipping
- • 2. Canines are often long, stout, and conical, and are usually used for catching, killing,
- and tearing prey. Canines and incisors have single roots
- • 3. Premolars are positioned next to canines, and have one or two roots and truncated
- surfaces for chewing
- • 4. Molars have broad chewing surfaces and two (upper molars) or three (lower molars)
- roots

• Mammalian species have characteristic numbers of each kind of adult tooth.

- **Qno35 Explain Characteristics of mammals?**
- • Any member of the group of vertebrate animals in which the young are nourished
- with milk from special mammary glands of the mother
- • In addition to these characteristic milk glands, mammals are distinguished by several
- other unique features. Hair is a typical mammalian feature, although in many whales it
- has disappeared except in the fetal stage
- • A chain of three tiny bones transmits sound waves across the middle ear
- • A muscular diaphragm separates the heart and the lungs from the abdominal cavity. Only
- the left aortic arch persists. (In birds the right aortic arch persists; in reptiles, amphibians,
- and fishes both arches are retained.)
- • Mature red blood cells (erythrocytes) in all mammals lack a nucleus; all other vertebrates
- have nucleated red blood cells
- • Mammalian characteristics evolved gradually over a 200-million-year period in the
- synapsid lineage.
- • Two subclasses of mammals evolved during the Mesozoic era—the Prototheria and the

- Theria. Modern mammals include monotremes, marsupial mammals, and placental
- mammals.
- • The skin of mammals is thick and protective and has an insulating covering of hair.
- • Adaptations of teeth and the digestive tract allow mammals to exploit a wide variety of
- food resources.
- • Efficient systems for circulation and gas exchange support the high metabolic rate
- associated with mammalian endothermy
- • The brain of mammals has an expanded cerebral cortex that processes information from
- various sensory structures.
- • Metanephric kidneys permit mammals to excrete urea without excessive water loss.
- • Complex behavior patterns enhance mammalian survival
- • Most mammals are viviparous and have reproductive cycles that help ensure internal
- fertilization and successful development.
- **Qno36 write a few lines on Hair?**
- • Hair is a keratinized derivative of the epidermis of the skin and is uniquely mammalian.
- It is seated in an invagination of the epidermis, called a hair follicle.
- • A coat of hair, called pelage, usually consists of two kinds of hair

- • Long guard hairs protect a dense coat of shorter, insulating under hairs. Because hair is

  composed largely of dead cells, it must be periodically molted.

- **Qno37 Explain Mammary Gland in mammals?**

- • Mammary glands are functional in female mammals and are present, but nonfunctional,
- in males
- • The milk that mammary glands secrete contains water, carbohydrates (especially the
- sugar lactose), fat, protein, minerals, and antibodies.
- • Mammary glands are probably derived evolutionarily from apocrine glands and usually
- contain substantial fatty deposits
- • Monotremes have mammary glands that lack nipples. The glands discharge milk into
- depressions on the belly, where the young lap it up
- In other mammals, mammary glands open via nipples or teats, and the young suckle for

  their nourishment.

- **Qno38 Explain External structure of reptiles?**

- ● Reptilian skin is thick, dry, and keratinized
- • Scales may be modified for various functions. For example, the large belly scales of
- snakes provide contact with the substrate during locomotion

- • Skin secretions include pheromones that function in sex recognition and defense
- • All reptiles periodically shed the outer, epidermal layers of the skin in a process called
- ecdysis
- • The frequency of ecdysis varies from one species to another, and it is greater in juveniles
- than adults
- • Cryptic coloration, mimicry, and aposematic coloration occur in reptiles. Color and color
- change also function in sex recognition and thermoregulation
- • Primitive reptiles move by crawling on earth surface
- • Many prehistoric reptiles were bipedal, meaning that they walked on hind limbs
- • They had a narrow pelvis and a heavy, outstretched tail for balance
- • Bipedal locomotion freed the front appendages, which became adapted for prey capture

  or flight in some animals.

- **Qno39 Explain Stomach of birds?**
- In many birds, a diverticulum of the esophagus, called the crop, is a storage structure that
- allows birds to quickly ingest large quantities of locally abundant food. They can then
- seek safety while digesting their meal

- The crop of pigeons produces "pigeon's milk," a cheesy secretion formed by the
- proliferation and sloughing of cells lining the crop. Young pigeons (squabs) feed on pigeon's milk until they are able to eat grain
- Cedar waxwings, vultures, and birds of prey use their esophagus for similar storage
- functions
- Crops are less well developed in insect-eating birds because insectivorous birds feed
- throughout the day on sparsely distributed food.
- The stomach of birds is modified into two regions. The proventriculus secretes gastric
- juices that initiate digestion (figure 21.9). The ventriculus (gizzard) has muscular walls
- to abrade and crush seeds or other hard materials
- Birds may swallow sand and other abrasives to aid digestion
- The bulk of enzymatic digestion and absorption occurs in the small intestine, aided by
- secretions from the pancreas and liver
- Paired ceca may be located at the union of the large and small intestine. These blind-
- ending sacs contain bacteria that aid in cellulose digestion.
- Birds usually eliminate undigested food through the cloaca; however, owls form pellets

- of bone, fur, and feathers that are ejected from the ventriculus through the mouth. Owl

  pellets accumulate in and around owl nests and are useful in studying their food habits.

Qno 40 What is the pineal body?

A pineal body on the roof of the forebrain appears to stimulate ovarian development and to regulate other functions influenced by light and dark periods.

Qno41 Give an example of urochordates?

Tunicates or sea squirts

- **Qno42 What is PLUMAGE?**

The covering of feathers on a bird is called the plumage.

- **Qno43 What is Asetivation?**

(Aestivation is a dormant state that helps an animal withstand hot, dry periods.) When rain again fills the lake or riverbed, lungfishes emerge from their burrows to feed and reproduce.

- **Qno44 What is ANNULI?**

This bird, Eoalulavis, was found in 115- million-year-old deposits and had a wingspan of 17 cm (roughly the same as a goldfinch). This fossil showed that Eoalulavis had a wing structure called an alula. Alula is present in many modern birds that engage in slow, hovering flight. Its presence in this fossil indicates that complex flight mechanisms associated with slow, hovering, and highly maneuverable flight evolved at least 115 million years ago.

- **Qno45 What is PLEGE?**

Pelage is reduced in large mammals from hot climates (e.g., elephants and hippopotamuses) and in some aquatic mammals (e.g., whales) that often have fatty insulation. A coat of hair, called pelage, usually consists of two kinds of hair • Long guard hairs protect a dense coat of shorter, insulating under hairs. Because hair is composed largely of dead cells, it must be periodically molted.

- **Qno46 Describe TYPES OF OPENINGS IN TEMPORAL REGIONS OF REPTILES?**

There are three classes of reptiles according to opening in temporal region • Reptiles in the subclass Anapsida lack openings or fenestrae region of the skull. The turtles represent this lineage today. • A second group of reptiles are diapsid. They have upper and lower openings in the temporal region of the skull. • Third group of reptiles are synapsid. They possess a single dorsal opening in the temporal region of the skull.

- **Qno47 Explain Altricial AND Precocial young's?**

- Some birds are helpless at hatching; others are more independent. Those that are entirely dependent on their parents are said to be altricial and they are often naked at hatching (figure 21.15a)
- • Altricial young must be brooded constantly at first because endothermy is not developed. They grow rapidly, and when they leave the nest, they are nearly as large as their parents. For example, American robins weigh 4 to 6 g at hatching and leave the nest 13 days later weighing 57 g

• Precocial young are alert and lively at hatching (figure 21.15b). They are usually covered with down and can walk, run, swim, and feed themselves—although one parent is usually present to lead the young to food and shelter FIGURE 21.15 Altricial and Precocial Chicks. (a) An

American robin (Turdus migratorius) feeding nestlings. Robins have altricial chicks that are helpless at hatching. (b) Killdeer (Charadrius vociferus) have precocial chicks that are down covered and can move.

- **Qno 48 What are Paddle fishes and from which structures it is derived?**

Paddlefishes are large, freshwater chondrostean. They have a large, paddlelike rostrum that is innervated with sensory organs believed to detect weak electrical fields (figure 18.13b). They swim through the water with their large mouths open, filtering crustaceans and small fishes. They are found mainly in lakes and large rivers of the Mississippi River basin and are also known from western North America and China. paddlefishes, and whale sharks are filter feeders. Long gill processes, called gill rakers, trap plankton while the fish is swimming through the water with its mouth open. One group of actinopterygians, the chondrosteans, contains many species that lived during the Permian, Triassic, and Jurassic periods (215 to 120 million years ago), but only 25 species remain today. Ancestral chondrosteans had a bony skeleton, but living members, the sturgeons and paddlefishes, have cartilaginous skeletons.

- **Qno49 Explain Mammary glands?**

Mammary glands are functional in female mammals and are present, but nonfunctional, in males • The milk that mammary glands secrete contains water, carbohydrates (especially the sugar lactose), fat, protein, minerals, and antibodies. • Mammary glands are probably derived evolutionarily from apocrine glands and usually contain substantial fatty deposits • Monotremes have mammary glands that lack nipples. The glands discharge milk into depressions on the belly, where the young lap it up • In other mammals, mammary glands open via nipples or teats, and the young suckle for their nourishment. Mammary Glands. Mammary glands

are specialized to secrete milk following the birth of young. (a) In humans, many ducts lead from the glands to a nipple. Parts of the duct system are enlarged to store milk. Suckling by an infant initiates a hormonal response that causes the mammary glands to release milk. (b) Some mammals (e.g., cattle) have teats that form by the extension of a collar of skin around the opening of mammary ducts. Milk collects in a large cistern prior to its release. The number of nipples or teats varies. Any member of the group of vertebrate animals in which the young are nourished with milk from special mammary glands of the mother.

Qno50 Explain orders of reptiles?

- They have four orders named as; 1. Testudines, 2. Rhynchocephalia 3. Squamata, 4. Crocodilia
- 1. Order testudines (Chelonia) • Members of the order Testudines or Chelonia are the turtles • The approximately 225 species of turtles are characterized by a bony shell, limbs articulating internally to the ribs, and a keratinized beak rather than teeth.
- 2. Order rhynchocephalia • The one surviving species of the order is the tuatara (Sphenodon punctatus)
- 3.Order squamata • Order squamata divided into three suborders.
- 1.Suborder Sauria—The Lizards
- 2. Suborder Serpentes—The Snakes
- 3.Suborder Amphisbaenia—Worm-Lizard

4. Order crocodilian • The order Crocodilia has 21 species • Along with dinosaurs, crocodilians are derived from the archosaurs Living crocodilians include the alligators, crocodiles, gavials, and caimans.

- **Qno51 What is ecdysis?**

Although reptilian skin is much less glandular than that of amphibians, secretions include pheromones that function in sex recognition and defense. All reptiles periodically shed the outer, epidermal layers of the skin in a process called ecdysis. Because the blood supply to the skin does not extend into the epidermis, the outer epidermal cells lose contact with the blood supply and die. Movement of lymph between the inner and outer epidermal layers loosens the outer epidermis. Ecdysis generally begins in the head region, and in snakes and many lizards, the epidermal layers come off in one piece. In other lizards, smaller pieces of skin flake off. The frequency of ecdysis varies from one species to another, and it is greater in juveniles than adults. Cryptic coloration, mimicry, and aposematic coloration occur in reptiles. Color and color change also function in sex recognition and thermoregulation.

- **Qno52 How lungs evolve in the fishes?**

The evolution of lungs in fishes was paralleled by changes in vertebrate circulatory systems. These changes are associated with the loss of gills, delivery of blood to the lungs, and separation of oxygenated and unoxygenated blood in the heart. Members of the subclass Sarcopterygii have muscular lobes associated with their fins and usually use lungs in gas exchange. One group of sarcopterygian gians are the lungfishes. Only three genera survive today, and all live in regions where seasonal droughts are common. When freshwater lakes and rivers begin to stagnate and dry, these fishes use lungs to breathe air. Some (Neo ceratodus) inhabit the freshwaters of Queensland, Australia. They survive stagnation by breathing air, but they normally use gills and cannot withstand total drying.

Mcqs

- **1-The allantois is**

- **a ventral outgrowth of the gut and stores nitrogenous wastes (e.g., uric acid).**
- **2-bony fishes possess outpockets of the intestine, called**
- **pyloric ceca, which increase absorptive and secretory surfaces**
- **3-The similarities between birds and reptiles are so striking that birds are often referred to as "glorified reptiles."**
- **4-plate of bone, called the secondary palate, evolved in the archosaurs and separates the nasal and mouth passage ways.**
- **5.Tail loss, or autotomy, is an adaptation that allows a lizard to escape from a predator's grasp, or the disconnected, wiggling piece of tail may distract a predator from the lizard.**
- **6-The tongues of turtles and crocodilians are non-protrusible and aid in swallowing.**
- **7-Most reptiles excrete uric acid. It is nontoxic, and being relatively insoluble in water, it precipitates in the excretory system.**
- **8-Members of the genus Lampetra are called brook lampreys.**
- **9-Eggs hatch in approximately three weeks into ammocoete larvae.**
- **10-a group of fishes called hag-fishes are the most primitive vertebrates known (living or extinct).**
- **11-It was present in the bony armor of a group of fishes called ostracoderms.**
- **12-In addition to their biting mouthparts and paired appendages, chondrichthians possess epidermal placoid scales and a cartilaginous endoskeleton.**

- **13-A third group of sarcopterygians, called osteolepiforms.**
- **14- Long gill processes, called gill rakers, trap plankton while the fish is swimming through the water with its mouth open.**
- **15- Reptiles in the subclass Anapsida lack openings or fenestrae region of the skull.**
- **16- The approximately 225 species of turtles are characterized by a bony shell, limbs, articulating internally to the ribs, and a keratinized beak rather than teeth.**
- **17- The dorsal portion of the shell is the carapace.**
- **18- The ventral portion of the shell is the plastron.**
- **19- Turtles have eight cervical vertebrae that can be articulated into an S-shaped.**
- **20- All turtles are oviparous they lay and cover with soil clutches of 5 to 100 eggs.**
- **21- About 3,300 species of lizards are in the suborder Sauria.**
- **22- Lizards vary in length from only a few centimeters to as large as 3 m.**
- **23- Geckos, commonly**
- **found on the walls of human dwellings in semitropical area.**
- **24- Iguanas have robust bodies, short necks, and distinct heads.**
- **25- Chameleons are adapted to arboreal lifestyles and use a long, sticky tongue to capture insects.**
- **26- About 2,300 species are in the suborder Serpentes.**

- **27- ● The skin of amphisbaenians has ringlike folds called annuli.**
- **28- The order Crocodilia has 21 species.**
- **29- Crocodilians have not changed much over their 170 million-year history.**
- **30- The crocodilians stomach is gizzardlike, and crocodilians swallow rocks and other objects as abrasives for breaking apart ingested food.**
- **Topic no48**
- **31-Reptilian skin is thick, dry, and keratinized**
- **32- All reptiles periodically shed the outer, epidermal layers of the skin in a process**
- **called ecdysis**
- **33- Tail loss, or autotomy, is an adaptation that allows a lizard to escape from a**
- **predator's grasp, or the disconnected, wiggling piece of tail may distract a predator**
- **from the lizard.**
- **34- Skull and jaws of reptiles help to capture large prey.**
- **35- Venom glands are modified salivary glands.**
- **36- ● In most reptiles, a negative-pressure mechanism is responsible for lung**
- **ventilation.**
- **37- To sustain activity, however, body temperatures are regulated within a narrow range,**

- **between 25 and 37° C.**
- **38- Heat loss from these groups, called hibernacula,.**
- **39- In snakes and some lizards, the upper and lower eyelids fuse in the embryo to**
- **form a protective window of clear skin, called the spectacle.**
- **40- A kidney with many more blood-filtering units, called nephrons**
- **41- The functional kidneys of adult reptiles are called metanephric kidneys.**
- **42- Lizards and snakes possess paired hemipenes at the base of the tail that are erected by being turned inside out, like the finger of a glove.**
- **43- Gonads lie in the abdominal cavity.**
- **44- Birds have horny beaks and scaly feet.**
- **45- Birds' complex mating systems and behavioral patterns increase the chances of offspring survival**

- **46- In 1861, one of the most important vertebrate fossils was found in a slate quarry in Bavaria, Germany.**
- **47- Sinornis fossils are 135 million years old—only 15 million years younger than Archaeopteryx.**
- **48- Eoalulavis, was found in 115- million-year-old deposits and had a wingspan of 17 cm (roughly the same as a goldfinch).**

- **49- . It is sufficient to say that adaptive radiation has resulted in about 9,100 species of living birds, which are divided into about 27 orders (table 21.1).**
- **50- The covering of feathers on a bird is called the plumage.**
- **51- Preening, which is done by rubbing the bill over the feathers, keeps the feathers smooth, clean,**
- **and in place.**
- **52- Birds also have a reduced number of skull bones, and a lighter, keratinized sheath called a bill replaces the teeth.**
- **53- The posterior caudal vertebrae are fused into a pygostyle, which helps support the tail feathers that are important in steering.**
- **54- The sternum of most birds bears a large, median keel for the attachment of flight muscles.**
- **55- Flapping flight generates the power for flight and is the most common type of flying.**
- **56- Hummingbirds perform hovering flight.**
- **57- The crop of pigeons produces "pigeon's milk**
- **58- The bird heart is relatively large (up to 2.4% of total body weight), and it beats rapidly. Rates in excess of 1,000 beats per minute have been recorded for hummingbirds under stress.**
- **59- The heart rate of an ostrich, for example, varies between 38 and 176 beats per minute.**
- **60- ● A special voice box, called the syrinx, The lungs of birds are made of small**

- **air tubes called parabronchi.**

- **61- Himalayas at altitudes of 9,200 m. A human mountain climber begins to feel the symptoms of altitude sickness and must carry auxiliary oxygen supplies above 2,500 m.**

Topic no 62

62 Birds maintain body temperatures between 38 and 45° C

63- Lethal extremes are lower than 32 and higher than 47° C

64- Shivering also generates heat in extreme cold

- **65-whippoorwills allow their body temperatures to drop from about 40° C to near 16° C, and respiratory rates become very slow.**
- **66-The corpus striatum functions in visual learning, feeding, courtship, and nesting.**
- **67-● A pineal body on the roof of the forebrain appears to stimulate ovarian development and to regulate other functions influenced by light and dark periods.**
- **68-The optic tectum (the roof of the mid brain), along with the corpus striatum, plays an important role in integrating sensory functions.**
- **69-Vision is an important sense for most birds.**
- **70-Cones are especially concentrated (1,000,000/mm2) at a focal point called the fovea.**

71-The one at the center of the retina is sometimes called the The one at the center of the retina is sometimes called the "search fovea" .

- **72-The posterior fovea is called the "pursuit fovea" because binocular vision produces depth perception, which is necessary to capture prey.**
- **73-Pigeons have eyes located well back on the sides of their head, giving them a nearly**
- **360° monocular field, but a narrow binocular field.**
- **74-Loose, delicate feathers called auriculars cover the external ear opening.**
- **75-The sensitivity of the avian ear (100 to 15,000 Hz) is similar to that of the human ear (16 to 20,000 Hz).**
- **76-Like reptiles, birds excrete uric acid, which is temporarily stored in the cloaca**
- **77-some birds have supraorbital salt glands that drain excess sodium chloride through the nasal openings to the outside of the body.**

78-All birds are oviparous.

79- Gonads are in the dorsal abdominal region, next to the kidneys.

- **80-In females, two ovaries form during development, but usually only the left ovary fully develops .**
- **81-Except for certain waterfowl and ostriches, birds have no intromittent organ.**

82-The oviduct opens into the cloaca.

83-Most birds are monogamous

84-Prairie chickens are polygynous, and males display in groups called leks

85-The group of eggs laid and chicks produced by a female is called a clutch.

86-The incubation period lasts between 10 and 80 days.

87-The Bird 🦅🕊️🐴that are dependent on their parents are said to be altricial.

88- American robins weigh 4 to 6 g at hatching and leave the nest 13 days later weighing 57 g.

- **89-Some long distance migrants may store fat equal to 50% of their body weight and make nonstop journeys.**
- **90-● In clear weather, many birds fly at altitudes greater than 1,000 m, which reduces the likelihood of hitting tall obstacles.**

91-Birds' lenses are transparent to ultraviolet light, and their photoreceptors respond to it, allowing them to orient using the sun, even on cloudy days. This orientation cue is called a sun compass.

- **92- The angle between the North Star and the horizon decreases as you move toward the equator.**
- **93-Any member of the group of vertebrate animals in which the young are nourished**
- **with milk from special mammary glands of the mother.**
- **94-Hair is a typical mammalian feature, although in**
- **many whales it has disappeared except in the fetal stage.**

95- A chain of three tiny bones transmits sound waves across the middle ear.

96- A muscular diaphragm separates the heart and the lungs from the abdominal cavity.

97-Mature red blood cells (erythrocytes) in all mammals lack a nucleus; all other vertebrates have nucleated red blood cells.

- **98-Mammalian characteristics evolved gradually over a 200-million-year period in the synapsid lineage.**
- **99-Two subclasses of mammals evolved during the Mesozoic era—the Prototheria and the Theria.**
- **100-The beginning of the Tertiary period, about 70 million years ago, was the start of the "age of mammals.**
- **101-By the middle of the Permian period , other successful mammal-like reptiles had arisen from the pelycosaurs. They were a diverse group known as the therapsids.**
- **102-● The last therapsids were a group called the cynodonts.**

103-● The first mammals were small (less than 10 cm long) with delicate skeletons.

104-The infraclass Ornithodelphia contains the monotremes.

105-The six species of monotremes are found in Australia and New Guinea.

- **106-About 250 species of marsupials live in the Australian region and the Americas.**
- **107-The approximately 3,800 species of eutherians are classified into 17 orders.**

- **108-Mammals are naturally distributed on all continents**
- **except Antarctica, and they live in all oceans.**

109-Hair is seated in an invagination of the epidermis, called a hair follicle.

110-A coat of hair, called pelage.

111- Guard hairs may sometimes be modified into thick-shafted hairs called vibrissae.

112-A band of smooth muscle, called the arrector pili muscle.

- **113-Arrector pili muscles are under the control of the**
- **autonomic nervous system, which also controls a mammal's "fight-or-flight" response.**
- **114-Pelage is reduced in large mammals from hot climates.**
- **115-Glands develop from the epidermis of the skin.**
- **116-Sebaceous (oil) glands are associated with hair follicles, and their oily secretion lubricates and waterproofs the skin and hair.**
- **117-Most mammals also possess sudoriferous (sweat) glands.**
- **118-Small sudoriferous glands (eccrine glands) release watery secretions used in evaporative cooling.**
- **119-Scent or musk glands are around the face, feet, or anus of many mammals.**
- **120-Larger sudoriferous glands (apocrine glands) secrete a mixture of salt, urea, and water, which microorganisms on the skin convert to odorous products.**
- **121-Mammary glands are functional in female mammals and are present, but nonfunctional, in males.**

122-● One feature that zoologists use to distinguish reptilian from mammalian skulls is the method of jaw articulation.

- **123-● In mammals, the secondary palate extends posteriorly by a fold of skin, called the soft palate.**

124-● In reptiles, the teeth are uniformly conical, a condition referred to as homodont.

125-The first teeth emerge before or shortly after birth and are called deciduous or milk teeth.

- **126-In mammals, the teeth are often specialized for different functions, a condition called heterodont.**

126-In rodents, the incisors grow throughout life.

127-The vertebral column of mammals is divided into five regions.

128-● A cecum is a fermentation pouch where microorganisms aid in cellulose digestion.

129-Sheep, cattle, and deer are called ruminants (L. ruminare, to chew the cud).

- **130-The evolution of similar structures in different lineages is called convergent evolution.**
- **131-Expiration is normally by elastic recoil of the lungs and relaxation of inspiratory muscles, which decreases the volume of the thoracic cavity.**
- **132-Shivering thermogenesis is muscular activity that generates large amounts of heat but little movement.**
- **133-walrus in cold, arctic waters has a surface temperature near 0° C; however, a fewcentimeters below the skin surface, body temperatures are about 35° C.**

- **134- most of the body of an arctic mammal is unusually well insulated.**
- **135-Countercurrent heat-exchange systems may help regulate heat loss from exposed areas .**
- **136-Hibernation is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates.**
- **137-Rodentia (e.g.,chipmunks and woodchucks), and Chiroptera (bats).**
- **138-After a hibernating mammal retreats to a burrow or a nest, the hypothalamus sets the body's thermostat to about 2° C.**
- **139-The respiratory rate of a hibernating ground squirrel falls from 100 to 200 breaths per minute to about four breaths per minute.**
- **140-The heart rate falls from 200 to 300 beats per minute to about 20 beats per minute.**
- **141-During hibernation, a mammal may lose a third to half of its body weight.**
- **142-The sensory patch of the inner ear that contains the sound receptors is long and coiled and is called the cochlea.**
- **143-Color vision is less well developed in mammals than in reptiles and birds.**

- **144-The primary adaptation of the mammalian nephron is a portion of the tubule system called the loop of the nephron.**

- **145-beavers produce urine that is twice as concentrated as blood, while**
- **Australian hopping mice produce urine that is 22 times more concentrated than Blood.**
- **146-In the nephron of the kidney, fluids and small solutes are filtered from the blood through the walls of a group of capillary like vessels, called the glomerulus.**

147-The primary adaptation of the mammalian nephron is a portion of the tubule system called the loop of the nephron.

148-Metabolic oxidation of carbohydrates produces water as a by-product.

149-Visual cues are often used in communication.

150-Vocalizations and tactile communication are important in primate social interactions.

- **151-Development is arrested for the three months during which the recently born young do most of their nursing. This mechanism is called embryonic ddiapause**
- **152-In viviparous mammals, the oviducts are modified into one or two uteri**
- **153-Many mammals are monestrus and have only a single yearly estrus cycle that is sharply seasonal.**
- **154-Rats and mice have estrus cycles that repeat every**
- **four to six days.**
- **155- In mammals Females store sperm in the uterus for periods in excess of two months.**

156-Female echidnas incubate eggs in a ventral pouch.

157-Platypus eggs are laid in their burrows.

- **158-The marsupial gestation period (the length of time young develop within the female reproductive tract) varies between**
- **8 and 40 days in different species.**
- **159-After birth, tiny young crawl into the marsupium and attach to a nipple, where they suckle for an additional 60 to 270 days**
- **160-Gestation periods of eutherian mammals vary from**

20 days (some rodents) to 19 months (the African elephant).

Finished and final part completed. Alhamdolillah with the grace of Allah Almighty I have completed my task with the hard work of many day and night.May Allah pak succeeded in my mission Aameen.

Iqbal group of bs zoology

03460004097-29/5/2022

Muhammad Iqbal arain ♡ Sanam Shah sahb


اللھم صل علی محمد وعلی آل محمد
شفاعت واجب ہوگئی
کریم آقا صلی اللہ علیہ وآلہ وسلم کا ارشاد گرامی ہے: جس نے یہ کہا
اَللّٰھُمَّ صَلِّ عَلٰی مُحَمَّدٍ وَّ اَنْزِلْهُ الْمَقْعَدَ
الْمُقَرَّبَ عِنْدَكَ يَوْمَ الْقِيَامَة
اس کے لیے میری شفاعت واجب ہوگئی۔ (معجم کبیر، جلد،
ص25، حدیث:448)

Muhammad Iqbal arain ♡ Nazia kanwal sahiba 03460004097



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وہ چند لمحے جوگزرے تیری رفاقت میں

اقبال

نہ جانے جانِ جاناں میرے کتنے برس کام آئے